

Question 1

A Generative AI Engineer is building a system for summarizing research papers. The papers often contain complex diagrams and tables. What should they prioritize when designing prompts?

1. Convert diagrams to text descriptions for summarization.

Explanation

Incorrect: Converting diagrams introduces noise and may distort the information.

2. Prioritize summarizing only abstract sections.

Explanation

Incorrect: Summarizing only abstracts reduces the depth and completeness of the output.

3. Emphasize summarizing text content only while ignoring diagrams and tables.

Explanation

Correct: Research papers contain structured text that is more suitable for LLM processing than visual content.

4. Include diagrams and tables directly in the prompt.

Explanation

Incorrect: LLMs do not process visual information, and including diagrams will not improve results.

Overall explanation

Prompting the LLM to focus on text ensures high-quality summaries without introducing noise from unsupported content like diagrams.

Question 2

A Generative AI Engineer must register an LLM in Unity Catalog using MLflow for better governance and reproducibility. What steps should they follow to register the model?

1. Train the model and directly upload it to Unity Catalog.

Explanation

Incorrect: Unity Catalog requires MLflow integration to manage metadata, versioning, and lineage effectively.

2. Log the trained model to MLflow, add the model signature, and register it in Unity Catalog for centralized management.

Explanation

Correct: Logging the model in MLflow, defining its signature, and registering it in Unity Catalog ensures governance, tracking, and reproducibility.

3. Fine-tune the model in Unity Catalog before deployment.

Explanation

Incorrect: Unity Catalog does not support model training or fine-tuning.

4. Define the input-output formats in Unity Catalog first, then register the model in MLflow.

Explanation

Incorrect: Model registration flows from MLflow to Unity Catalog, not the reverse.

Overall explanation

Registering an LLM in Unity Catalog involves logging the model in MLflow, defining its input-output signature, and integrating it for governance and management.

Question 3

A Generative AI Engineer needs to serve a Retrieval-Augmented Generation (RAG) application that uses a foundation model API. What is the key consideration for cost optimization?

1. Remove retrievers to simplify the pipeline.

Explanation

Incorrect: Retrievers are essential for RAG workflows and cannot be excluded.

2. Enable real-time query logging for monitoring.

Explanation

Incorrect: Logging helps with debugging but does not reduce costs.

3. Batch multiple user queries during inference to reduce API calls.

Explanation

Correct: Batching queries reduces the number of API calls, lowering costs without compromising performance.

4. Use a larger model to improve accuracy.

Explanation

Incorrect: Larger models may improve performance but increase costs significantly.

Overall explanation

Batching queries reduces API call frequency, which helps optimize costs while maintaining system performance.

Question 4

A Generative AI Engineer is tasked with building a pipeline for retrieving technical documents and answering user queries. Which strategy best ensures accurate responses?

1. Focus on using the smallest LLM available.

Explanation

Incorrect: Smaller models may lack the capabilities to handle complex queries effectively.

2. Rely on generic prompts that do not include document-specific details.

Explanation

Incorrect: Generic prompts reduce accuracy by excluding critical context.

3. Use only pre-trained embeddings for document matching.

Explanation

Incorrect: Embeddings are helpful but do not provide the required context for detailed responses.

4. Combine document retrieval with context-based prompt augmentation to include relevant details in user queries.

Explanation

Correct: Augmenting user prompts with retrieved context improves the LLM's ability to generate accurate responses.

Overall explanation

Combining document retrieval with context augmentation ensures accurate and context-aware responses tailored to user queries.

Question 5

A Generative AI Engineer is evaluating the performance of a RAG application using MLflow. What metrics should they track to assess retrieval effectiveness?

1. BLEU and ROUGE scores for evaluating text generation.

Explanation

Incorrect: BLEU and ROUGE assess language generation, not retrieval effectiveness.

2. Token usage per query.

Explanation

Incorrect: Token usage measures cost efficiency, not retrieval quality.

3. Precision, recall, and NDCG (Normalized Discounted Cumulative Gain).

Explanation

Correct: These metrics evaluate retrieval quality, with precision measuring relevance, recall capturing completeness, and NDCG assessing ranking accuracy.

4. Model perplexity during training.

Explanation

Incorrect: Perplexity measures language model fluency but does not evaluate retrieval.

Overall explanation

Tracking precision, recall, and NDCG ensures a comprehensive evaluation of retrieval performance in a RAG application.

Question 6

A Generative AI Engineer just deployed an LLM application at a digital marketing company that assists with answering customer service inquiries. Which metric should they monitor to evaluate the application's production performance?

1. HuggingFace Leaderboard values for the base LLM.

Explanation

Incorrect: These values assess general model benchmarks but do not indicate the deployed application's performance.

2. Number of customer inquiries processed per unit of time.

Explanation

Correct: Monitoring throughput helps ensure the application can handle the expected volume of customer queries efficiently.

3. Energy usage per query.

Explanation

Incorrect: While useful for cost optimization, this metric does not assess application performance.

4. Final perplexity scores for the training of the model.

Explanation

Incorrect: Perplexity is relevant for evaluating training, not production performance.

Overall explanation

Throughput monitoring is critical in production environments to measure how effectively the application handles customer inquiries.

Question 7

A Generative AI Engineer is developing a RAG application to answer questions about financial regulations. The retrieved content often contains irrelevant sections. What approach can improve the retrieval process?

1. Implement an embedding model fine-tuned on financial text to improve semantic relevance.

Explanation

Correct: Fine-tuning embeddings ensures better alignment with financial regulation queries.

2. Rely on a rule-based filtering system.

Explanation

Incorrect: Rule-based systems lack flexibility and adaptability for diverse queries.

3. Use a summarization model to reduce irrelevant content.

Explanation

Incorrect: Summarization does not address the retrieval process directly.

4. Increase the retrieval batch size to include more documents.

Explanation

Incorrect: Larger batch sizes may increase noise without improving relevance.

Overall explanation

Fine-tuning an embedding model on domain-specific text improves retrieval relevance by aligning with the context of financial regulations.

Question 8

A Generative AI Engineer must clean a dataset of news articles for a RAG system. The articles include unrelated advertisements and redundant metadata. What preprocessing step is necessary?

1. Retain all parts of the dataset to preserve the original structure.

Explanation

Incorrect: Retaining extraneous content reduces retrieval relevance and increases noise.

2. Summarize each article into a shorter format.

Explanation

Incorrect: Summarization does not address the issue of irrelevant metadata or advertisements.

3. Remove advertisements and metadata to retain only the main article content.

Explanation

Correct: Cleaning irrelevant content ensures the RAG system retrieves meaningful information efficiently.

4. Merge all articles into a single file for consistency.

Explanation

Incorrect: Merging does not resolve the issue of extraneous content.

Overall explanation

Removing irrelevant content like advertisements and metadata ensures the dataset is clean, improving retrieval and query accuracy.

Question 9

A Generative AI Engineer is building a pipeline to deploy an LLM that generates personalized travel itineraries. The model needs to handle location data, user preferences, and budget constraints. What key components should the pipeline include?

1. An embedding model for location and user preference representation, a retriever for querying data, and an LLM for generating itineraries.

Explanation

Correct: These components work together to dynamically process inputs and generate tailored itineraries.

2. A summarization model to condense travel options.

Explanation

Incorrect: Summarization reduces detail, which is critical for personalized itineraries.

3. A rule-based system for fixed travel recommendations.

Explanation

Incorrect: Rule-based systems lack the flexibility needed for dynamic inputs.

4. A classification model to categorize user preferences.

Explanation

Incorrect: Classification is insufficient for generating comprehensive, personalized itineraries.

Overall explanation

Embedding models, retrievers, and LLMs combine to generate accurate and personalized travel plans based on diverse user inputs.

Question 10

After changing the response-generating LLM in a RAG pipeline from GPT-4 to a model with a shorter context length that the company self-hosts, the Generative AI Engineer is getting the following error: {"error_code": "BAD_REQUEST", "message": "Bad request: rpc error: code = InvalidArgument desc = prompt token count (4595) cannot exceed 4096..."} What TWO solutions should the Generative AI Engineer implement without changing the response-generating model? **(Choose two)**

1. Reduce the number of records retrieved from the vector database

Explanation

Correct: Limiting the number of retrieved records reduces the total token count, effectively addressing the issue.

2. Reduce the maximum output tokens of the new model

Explanation

Incorrect: While this limits the output token count, it doesn't address excessive input tokens from the vector database.

3. Use a smaller embedding model to generate embeddings

Explanation

Incorrect: The error is related to token count exceeding the model's context length, not the size of the embedding model.

4. Decrease the chunk size of embedded documents

Explanation

Correct: Reducing the chunk size ensures fewer tokens are retrieved from each document, helping to stay within the model's context limit.

5. Retrain the response-generating model using ALiBi

Explanation

Incorrect: Retraining the model is unnecessary and overly complex for solving a token limit issue.

Overall explanation

By decreasing the chunk size of embedded documents and reducing the number of retrieved records, the engineer can ensure the total token count remains within the model's context length without requiring significant changes to the system.

Question 11

A Generative AI Engineer must deploy a secure endpoint for a RAG application. The endpoint handles sensitive legal documents and must be compliant with data privacy regulations. What measures are essential? **(Choose two)**

1. Skip authentication to allow unrestricted access.

Explanation

Incorrect: Authentication is critical for securing sensitive systems.

2. Implement API key-based authentication for endpoint access.

Explanation

Correct: API key authentication ensures only authorized users can access the endpoint.

3. Use public-facing endpoints without encryption to simplify access.

Explanation

Incorrect: Public endpoints without encryption expose sensitive data to security risks.

4. Encrypt data at rest and in transit to protect sensitive information.

Explanation

Correct: Encryption ensures data security and compliance with privacy regulations.

5. Rely on a shared access key for all users.

Explanation

Incorrect: Shared keys are insecure and lack user-specific controls.

Overall explanation

Encryption and API key authentication are critical measures for secure and compliant endpoint deployment.

Question 12

A Generative AI Engineer must create a pipeline to retrieve and summarize customer reviews. Some reviews contain offensive language. How should they handle preprocessing?

1. Summarize without filtering.

Explanation

Incorrect: Summarization without filtering may include offensive language in the output.

2. Implement filtering to detect and remove offensive language before summarization.

Explanation

Correct: Filtering offensive content ensures summaries remain safe and appropriate for end users.

3. Retain offensive language for context.

Explanation

Incorrect: Offensive content can harm the user experience and is unnecessary for summaries.

4. Exclude all reviews containing more than 10 words.

Explanation

Incorrect: Excluding reviews based on length removes relevant content.

Overall explanation

Filtering offensive language ensures that summaries remain appropriate and align with user expectations.

Question 13

A Generative AI Engineer needs to implement a safety guardrail for a chatbot that handles medical inquiries. The chatbot must avoid giving medical advice but still provide relevant information. What guardrail technique is most appropriate?

1. Train the LLM on a dataset with neutral language.

Explanation

Incorrect: Training alone cannot ensure the avoidance of medical advice in responses.

2. Reduce the model temperature to limit randomness.

Explanation

Incorrect: Temperature adjustments reduce variability but do not address safety concerns.

3. Apply a classifier to flag risky outputs post-generation.

Explanation

Incorrect: Post-generation classifiers are less efficient than proactive metaprompting.

4. Use metaprompts instructing the chatbot to defer medical advice and only provide general information.

Explanation

Correct: Metaprompts guide the chatbot's behavior, ensuring it avoids giving medical advice while providing relevant details.

Overall explanation

Metaprompts provide a proactive guardrail, ensuring the chatbot complies with safety and ethical guidelines.

Question 14

A Generative AI Engineer must register a deployed model to Unity Catalog using MLflow. What is the correct workflow?

1. Skip configuring metadata during registration.

Explanation

Incorrect: Metadata provides critical information for model management and discoverability.

2. Deploy the model directly to Unity Catalog without logging it to MLflow.

Explanation

Incorrect: MLflow logging is essential for proper metadata and tracking before registration.

3. Train the model, log it to MLflow, configure metadata, and register it to Unity Catalog.

Explanation

Correct: This workflow ensures that the model is logged, annotated, and properly integrated with Unity Catalog for discoverability.

4. Log the model to MLflow but skip registering it to Unity Catalog.

Explanation

Incorrect: Unity Catalog registration ensures centralized governance and access control.

Overall explanation

MLflow provides seamless logging and metadata configuration, while Unity Catalog ensures model discoverability and governance.

Question 15

Why is it essential to evaluate an LLM's context length when integrating it into a document summarization pipeline?

1. To reduce token usage and inference cost.

Explanation

Incorrect: While context length affects cost, its primary role is handling input effectively.

2. To ensure the model can process input documents without truncation, preserving key information.

Explanation

Correct: Adequate context length ensures the entire document is processed, avoiding loss of critical details.

3. To ensure shorter summaries.

Explanation

Incorrect: Context length does not directly control summary length.

4. To improve model training efficiency.

Explanation

Incorrect: Context length is unrelated to training efficiency; it impacts input handling during inference.

Overall explanation

Evaluating context length ensures the model can process the full document without losing critical details due to truncation.

Question 16

A Generative AI Engineer must design a pipeline for generating personalized travel itineraries. The system must incorporate budget constraints, user preferences, and destination details. What components are critical? (Choose two)

1. Use a classification model to categorize destinations by type.

Explanation

Incorrect: Classification aids categorization but does not generate itineraries.

2. Use an embedding model to represent destinations and preferences semantically.

Explanation

Correct: Embeddings enable contextual understanding of user preferences and destinations.

3. Use an LLM to generate detailed itineraries dynamically.

Explanation

Correct: LLMs tailor travel plans based on user inputs, incorporating preferences and constraints.

4. Use a summarization model to condense user inputs into key points.

Explanation

Incorrect: Summarization does not create detailed itineraries.

5. Use a rule-based system to match users with predefined itineraries.

Explanation

Incorrect: Rule-based systems lack flexibility for dynamic personalization.

Overall explanation

Combining embeddings and LLMs ensures dynamic, contextually accurate travel itineraries tailored to user needs.

Question 17

A Generative AI Engineer must evaluate the retrieval performance of a chatbot. The system retrieves many irrelevant documents but also misses some relevant ones. Which metrics should they focus on? (Choose four)

1. F1 Score

Explanation

Correct: The F1 Score balances precision and recall, providing a holistic view of retrieval performance.

2. Precision

Explanation

Correct: Precision measures the relevance of retrieved documents, helping reduce irrelevant outputs.

3. Sentiment Analysis

Explanation

Incorrect: Sentiment analysis evaluates tone but is unrelated to retrieval performance.

4. Recall

Explanation

Correct: Recall ensures the system retrieves all relevant documents, reducing missed information.

5. Cosine Similarity

Explanation

Correct: Cosine similarity assesses semantic alignment, helping identify the relevance of retrieved content.

Overall explanation

Focusing on precision, recall, F1 score, and cosine similarity ensures a comprehensive evaluation of retrieval performance, addressing both relevance and completeness.

Question 18

A Generative AI Engineer must implement masking techniques to handle personally identifiable information (PII) in a customer feedback system. What is the primary advantage of this approach?

1. Masking reduces the response time for processing user data.

Explanation

Incorrect: Response time is not directly impacted by masking techniques.

2. Masking prevents model overfitting during training.

Explanation

Incorrect: Masking focuses on privacy, not training optimization.

3. Masking encrypts the dataset to protect it during storage.

Explanation

Incorrect: Encryption and masking serve different purposes; masking anonymizes data, while encryption secures it.

4. Masking anonymizes sensitive data, reducing privacy risks while preserving analytical value for feedback processing.

Explanation

Correct: Masking techniques replace sensitive information with placeholders, ensuring privacy without losing context.

Overall explanation

Masking protects sensitive data by anonymizing PII, ensuring privacy compliance and preserving the usefulness of customer feedback data.

Question 19

A Generative AI Engineer must prioritize high-quality source documents for training a financial RAG system. Which sources should they focus on?

1. User-written financial blogs.

Explanation

Incorrect: Blogs may contain inaccuracies and personal opinions, making them unreliable for training.

2. Regulatory filings and official financial reports.

Explanation

Correct: These sources are authoritative and ensure accurate and reliable training data.

3. User-submitted financial forum posts.

Explanation

Incorrect: Forums often contain unverified or biased content unsuitable for training.

4. News articles summarizing financial events.

Explanation

Incorrect: News articles are secondary sources and may lack detail.

Overall explanation

Using authoritative sources like regulatory filings ensures accurate, trustworthy data for financial RAG applications.

Question 20

A Generative AI Engineer is designing a chain using LangChain to answer user questions about technical specifications of electronic devices. The chain must handle multi-turn conversations and fetch detailed responses. What elements should the chain include? **(Choose two)**

1. A sentiment analysis model to detect user satisfaction.

Explanation

Incorrect: Sentiment analysis is not relevant to fetching technical specifications.

2. A summarization model to shorten long responses.

Explanation

Incorrect: Summarization is unnecessary for detailed responses required in this context.

3. A prompt template that maintains conversation history for context.

Explanation

Correct: Maintaining conversation history ensures the chain generates coherent multi-turn responses.

4. A retriever to fetch device specifications from a vector store.

Explanation

Correct: Retrievers enable access to relevant technical data for user queries.

5. A rule-based system to provide fixed answers for repetitive queries.

Explanation

Incorrect: Rule-based systems lack flexibility for dynamic multi-turn conversations.

Overall explanation

Prompt templates and retrievers ensure the LangChain application provides context-aware and detailed responses for technical queries.

Question 21

Which temperature value setting encourages a large language model to generate creative and varied responses?

1. Lower the temperature to a value close to 0.

Explanation

Incorrect: Lower temperatures reduce randomness, leading to more deterministic and less creative outputs.

2. Set the temperature to exactly 0.5.

Explanation

Incorrect: A mid-range temperature balances randomness and determinism but does not maximize creativity.

3. Disable temperature controls for unconstrained outputs.

Explanation

Incorrect: Most models require a temperature setting to balance creativity and coherence.

Correct answer

4. Increase the temperature to a higher value, like 1.0 or above.

Explanation

Correct: Higher temperatures introduce more randomness, enabling creative and diverse outputs suitable for open-ended tasks.

Overall explanation

Increasing the temperature encourages randomness, enabling the model to generate varied, creative responses ideal for brainstorming or storytelling tasks.

Question 22

A Generative AI Engineer is tasked with filtering irrelevant sections from product manuals used in a RAG system. The manuals include marketing descriptions, disclaimers, and technical specifications. Which sections should be retained?

1. Remove all sections and rely solely on pre-trained model knowledge.

Explanation

Incorrect: Removing all sections eliminates critical document context for retrieval.

2. Retain only the technical specifications relevant to user queries.

Explanation

Correct: Focusing on technical sections aligns with user needs and improves retrieval relevance.

3. Retain all sections for completeness.

Explanation

Incorrect: Irrelevant sections like marketing descriptions add noise and degrade system performance.

4. Include disclaimers and technical specifications.

Explanation

Incorrect: Disclaimers are rarely relevant for query resolution.

Overall explanation

Filtering irrelevant sections like marketing descriptions ensures the RAG system retrieves precise, contextually relevant information.

Question 23

A Generative AI Engineer must clean a dataset of scanned financial documents containing extraneous information like watermarks and logos. What preprocessing step is most effective?

1. Use image preprocessing techniques to filter out watermarks and logos before text extraction.

Explanation

Correct: Preprocessing images removes visual noise, improving text extraction quality.

2. Apply a summarization model to condense extracted content.

Explanation

Incorrect: Summarization condenses text but does not address cleaning.

3. Remove sections with missing text.

Explanation

Incorrect: Removing incomplete sections does not target extraneous content.

4. Retain all data to preserve document authenticity.

Explanation

Incorrect: Extraneous elements degrade the quality of extracted text and downstream tasks.

Overall explanation

Image preprocessing ensures cleaner text extraction, improving the quality of downstream tasks.

Question 24

A Generative AI Engineer is monitoring a deployed RAG system that shows a decline in user satisfaction due to irrelevant outputs. The retrieval system has high latency. What metrics should the engineer focus on to diagnose and resolve the issue?

1. Token usage and response length to optimize costs.

Explanation

Incorrect: These metrics do not address relevance or latency issues in retrieval.

2. Perplexity and BLEU to evaluate output fluency and accuracy.

Explanation

Incorrect: These metrics focus on language generation, not retrieval performance.

3. User query frequency to identify demand patterns.

Explanation

Incorrect: Query frequency insights do not directly address relevance or performance issues.

4. Precision, recall, and retrieval latency to measure relevance and identify bottlenecks.

Explanation

Correct: Precision and recall assess relevance, while retrieval latency highlights performance bottlenecks in the retrieval step.

Overall explanation

Precision, recall, and latency provide critical insights into relevance and retrieval efficiency, helping resolve issues with irrelevant outputs.

Question 25

A large e-commerce platform needs a product recommendation engine that generates personalized suggestions using user purchase histories. Which pipeline design best fits this use case?

1. Generate recommendations based on randomly sampled products.

Explanation

Incorrect: Random sampling does not align with user preferences or needs.

2. Use embeddings for purchase history and generate recommendations based on similarity scores.

Explanation

Correct: Embedding-based similarity ensures personalized and contextually relevant recommendations.

3. Use rule-based filters for predefined product categories.

Explanation

Incorrect: Rule-based filters lack the flexibility for personalized recommendations.

4. Implement a purely generative approach without historical data.

Explanation

Incorrect: Generative models alone cannot provide personalized recommendations without context.

Overall explanation

Embedding-based pipelines are ideal for personalized recommendations as they account for user preferences and purchase histories.

Question 26

A Generative AI Engineer needs to extract text from scanned invoices in JPG format for a financial application. Which Python package is best suited for this task?

1. PyPDF2 for extracting text from images.

Explanation

Incorrect: PyPDF2 is designed for PDFs, not image files.

2. BeautifulSoup for parsing text.

Explanation

Incorrect: BeautifulSoup is used for HTML and XML parsing, not image-based OCR tasks.

3. pdfplumber to parse the images.

Explanation

Incorrect: pdfplumber is designed for extracting text from PDF documents, not images.

4. Tesseract OCR via the pytesseract package.

Explanation

Correct: pytesseract is widely used for text extraction from scanned images, including JPG formats.

Overall explanation

pytesseract enables robust OCR for extracting text from image files like JPGs, making it ideal for this task.

Question 27

A Generative AI Engineer is selecting an LLM for a RAG application. The application retrieves long documents, and the model must summarize key sections accurately. What factors should guide the LLM choice?

1. Retrieval precision and recall.

Explanation

Incorrect: Retrieval metrics evaluate document retrieval quality, not LLM summarization performance.

2. Token limit and BLEU scores.

Explanation

Incorrect: BLEU scores assess text generation but do not address summarization quality or inference speed.

3. Model size and training dataset diversity.

Explanation

Incorrect: While important, these factors do not directly align with the specific application requirements.

4. Context length, summarization quality, and inference speed.

Explanation

Correct: These factors ensure the LLM can process long documents, summarize effectively, and deliver timely responses.

Overall explanation

Context length, summarization quality, and inference speed ensure the LLM is suitable for long-document summarization in a RAG application.

Question 28

A Generative AI Engineer is developing a chatbot designed to assist users with insurance-related queries. To comply with company policy, the chatbot must avoid responding to political inquiries and instead provide a standard message: "Sorry, I cannot answer that. I am a chatbot that can only answer questions around insurance." Which framework type should be implemented to solve this?

1. Security Guardrail

Explanation

Incorrect: Security guardrails focus on protecting sensitive data and infrastructure, not output filtering.

2. Compliance Guardrail

Explanation

Incorrect: Compliance guardrails enforce adherence to legal and regulatory standards but are not specific to restricting political responses.

3. Safety Guardrail

Explanation

Correct: A safety guardrail ensures that inappropriate or undesired outputs, such as political responses, are filtered or replaced with a predefined message.

4. Contextual Guardrail

Explanation

Incorrect: Contextual guardrails adjust model outputs based on user context but do not filter specific types of questions.

Overall explanation

Safety guardrails are specifically designed to block or modify inappropriate outputs, ensuring the chatbot aligns with company policy by restricting responses to political inquiries.

Question 29

A Generative AI Engineer is building a conversational agent for a banking application. The agent must handle sensitive information securely. What is the best approach to ensure data security?

1. Use summarization models to reduce data volume.

Explanation

Incorrect: Summarization does not inherently secure sensitive data.

2. Implement encryption for sensitive data and use LLM guardrails to avoid leakage.

Explanation

Correct: Combining encryption and guardrails ensures secure handling of sensitive information in conversational contexts.

3. Train the LLM on datasets with anonymized data.

Explanation

Incorrect: Anonymized datasets alone do not prevent data leakage in outputs.

4. Lower the temperature to limit variability in responses.

Explanation

Incorrect: Temperature adjustments do not address data security concerns.

Overall explanation

Encryption combined with LLM guardrails provides robust security for handling sensitive data in banking applications.

Question 30

A Generative AI Engineer must implement guardrails for a chatbot handling legal inquiries to prevent generating inaccurate legal advice. What technique is most effective? **(Choose two)**

1. Lower the model temperature to limit output variability.

Explanation

Incorrect: Temperature adjustments reduce randomness but do not address the core issue of inaccurate advice.

2. Block user queries containing legal terms.

Explanation

Incorrect: Blocking queries reduces usability and limits the chatbot's functionality.

3. Use metaprompts to instruct the chatbot to limit responses to general information, avoiding legal advice.

Explanation

Correct: Metaprompts guide the model's behavior, reducing the risk of generating inaccurate or inappropriate responses.

4. Train the chatbot on legal datasets to improve accuracy.

Explanation

Incorrect: Training improves knowledge but does not guarantee accurate or appropriate outputs.

5. Add a disclaimer in all chatbot outputs clarifying its limitations.

Explanation

Correct: Disclaimers set user expectations, minimizing the risk of misuse.

Overall explanation

Combining metaprompts with disclaimers ensures user safety while maintaining the chatbot's utility.

Question 31

A Generative AI Engineer must identify resources required to serve features for a RAG application deployed at scale. What resources are critical? **(Choose two)**

1. Sufficient compute power for embeddings and retrieval, storage for vector data, and monitoring tools for performance.

Explanation

Correct: Compute, storage, and monitoring are vital for scaling RAG applications efficiently.

2. Additional LLM fine-tuning capabilities.

Explanation

Incorrect: Fine-tuning is not critical for serving features at scale.

3. Tokenization optimization tools.

Explanation

Incorrect: Tokenization is a preprocessing task, not a serving resource.

4. Batch processing tools for low-latency responses.

Explanation

Incorrect: Batch processing increases latency, unsuitable for real-time RAG applications.

5. High-speed internet connections to minimize latency.

Explanation

Correct: Network speed affects real-time retrieval and response performance.

Overall explanation

Compute power, storage, monitoring, and high-speed networks ensure efficient, scalable RAG application serving.

Question 32

A Generative AI Engineer is tasked with designing an AI pipeline for translating unstructured user feedback into actionable tasks. The pipeline must highlight priorities and deadlines from the feedback. What is the best approach?

1. Use a classification model to categorize feedback.

Explanation

Incorrect: Categorization alone does not produce actionable tasks.

2. Use an LLM to analyse user feedback and generate structured outputs with priorities and deadlines.

Explanation

Correct: LLMs process unstructured input and generate actionable, structured outputs effectively.

3. Use a rule-based system to extract keywords.

Explanation

Incorrect: Rule-based systems lack flexibility and adaptability for diverse feedback.

4. Use a summarization model to condense feedback.

Explanation

Incorrect: Summarization does not extract priorities or deadlines.

Overall explanation

LLMs dynamically translate unstructured feedback into structured, actionable outputs, meeting business requirements.

Question 33

A Generative AI Engineer must implement guardrails to prevent a generative model from leaking sensitive data in its responses. What is the most effective strategy?

1. Use post-processing filters to remove sensitive information.

Explanation

Incorrect: Post-processing may catch some issues but is less proactive than metaprompts.

2. Use metaprompts to explicitly instruct the model not to include sensitive data.

Explanation

Correct: Metaprompts are effective in guiding model behavior and preventing data leakage.

3. Apply a summarization model to shorten responses.

Explanation

Incorrect: Summarization does not directly prevent sensitive data leakage.

4. Lower the model temperature to reduce randomness.

Explanation

Incorrect: Lower temperature reduces variability but does not prevent data leakage.

Overall explanation

Metaprompts ensure proactive control over the model's output, preventing sensitive data leakage effectively.

Question 34

A Generative AI Engineer is tasked with designing an assistant for a financial services company. The assistant must recommend personalized savings plans based on user income, expenses, and goals. What should the prompt include? **(Choose two)**

1. Use a generic prompt like "Generate a savings plan."

Explanation

Incorrect: Generic prompts may lack sufficient detail, leading to vague or irrelevant responses.

2. Provide an example of a successful savings plan.

Explanation

Correct: Examples help guide the model to align its output with practical financial strategies.

3. Specify input details like income, expenses, and goals, and request an actionable recommendation.

Example: "Income: \$5000, Expenses: \$3000. Goal: Save for a vacation. Provide a savings recommendation."

Explanation

Correct: Providing specific inputs and clear output expectations ensures the model generates actionable and relevant savings plans.

4. Avoid specifying goals to encourage flexibility.

Explanation

Incorrect: Goals are essential for generating personalized recommendations.

Overall explanation

Specifying inputs (income, expenses, goals) and including examples ensures the model generates personalized, actionable savings recommendations aligned with user needs.

Question 35

A Generative AI Engineer is integrating a third-party dataset into a RAG application. Some documents in the dataset are flagged for containing harmful misinformation. What mitigation strategy should they use?

1. Retain flagged documents but monitor their impact during inference.

Explanation

Incorrect: Monitoring does not prevent misinformation from affecting application outputs.

2. Use the flagged documents but add disclaimers to outputs.

Explanation

Incorrect: Disclaimers do not mitigate the risk of misinformation affecting the system.

3. Exclude flagged documents during preprocessing to prevent misinformation from entering the application.

Explanation

Correct: Excluding flagged documents ensures the dataset remains reliable and free from harmful misinformation.

4. Allow flagged documents but train the model to ignore them.

Explanation

Incorrect: Models cannot reliably exclude flagged content without preprocessing.

Overall explanation

Excluding flagged documents ensures the application avoids misinformation, maintaining reliability and ethical standards.

Question 36

A Generative AI Engineer is building a multi-stage reasoning system to assist users in legal contract reviews. The system should identify potential issues, propose revisions, and highlight key clauses. What tools are essential? (Choose two)

1. A visualization tool to display highlighted clauses.

Explanation

A visualization tool for displaying highlighted clauses may enhance the user experience, but it is not as essential as the embedding model and retriever for the core functionality of the multi-stage reasoning system in legal contract reviews. This tool may improve the presentation of results but is not a fundamental requirement for the system's functionality.

2. A classification model to label clauses by type.

Explanation

While a classification model can be useful for labeling clauses by type, it is not as essential as the embedding model and retriever for the multi-stage reasoning system required for legal contract reviews. This tool may enhance the system's capabilities but is not a core requirement.

3. A summarization model to condense contract text.

Explanation

A summarization model can help condense contract text, but it is not as crucial as the embedding model and retriever for the multi-stage reasoning system needed for legal contract reviews. While summarization can improve readability, it is not as vital for identifying potential issues and proposing revisions.

4. A retriever to fetch relevant legal references and precedents.

Explanation

A retriever is necessary to fetch relevant legal references and precedents that can be used to support the reasoning process in legal contract reviews. This tool helps the system provide accurate and well-informed suggestions to users.

5. An embedding model to semantically analyse contract clauses.

Explanation

An embedding model is essential for semantically analyzing contract clauses, allowing the system to understand the meaning and context of the text. This is crucial for identifying potential issues and proposing revisions in legal contract reviews.

Overall explanation

Combining embeddings and retrievers ensures effective issue identification and revision proposals for legal contracts.

Question 37

A Generative AI Engineer is tasked with querying a large vector database for a knowledge management system. How should they optimize the query process for performance and accuracy?

1. Perform brute-force search for maximum precision.

Explanation

Incorrect: Brute-force search is computationally expensive and impractical for large datasets.

2. Use nearest-neighbor search algorithms with appropriate distance metrics for accurate retrieval.

Explanation

Correct: Nearest-neighbor search algorithms optimize performance and accuracy in vector database queries.

3. Reduce query complexity by using smaller embedding models.

Explanation

Incorrect: Smaller embeddings may degrade semantic accuracy.

4. Skip embedding optimization and rely on raw text retrieval.

Explanation

Incorrect: Raw text retrieval does not leverage the power of vector search for semantic matching.

Overall explanation

Nearest-neighbor search with optimized distance metrics ensures high-performance, accurate vector database querying.

Question 38

A team is building a RAG system for answering technical queries. Despite having a robust retriever, the system's responses lack specificity. How can they improve the quality of answers?

1. Fine-tune the retriever for classification tasks.

Explanation

Incorrect: Classification tasks do not directly enhance answer specificity.

2. Use a smaller LLM to focus on brevity.

Explanation

Incorrect: Smaller models may compromise response quality rather than improve specificity.

3. Augment the retriever with a context-aware summarizer to provide more precise and relevant answers.

Explanation

Correct: Summarization models refine retrieved content, ensuring specific and targeted answers to technical queries.

4. Increase the size of the retriever index.

Explanation

Incorrect: A larger index increases retrieval options but does not guarantee answer quality.

Overall explanation

Combining retrieval with context-aware summarization ensures that responses are specific and relevant, improving the overall quality of answers.

Question 39

A Generative AI Engineer is selecting a model for generating summaries of financial reports. The reports often contain tables with numerical data. What should they prioritize?

1. Use a generative model trained on open-domain text.

Explanation

Incorrect: Open-domain training does not guarantee domain-specific performance.

2. Choose a model optimized for numerical reasoning and summarization tasks.

Explanation

Correct: Models with strong numerical reasoning handle data-rich content more effectively.

3. Focus on the model with the largest context length.

Explanation

Incorrect: While context length is helpful, numerical reasoning is more critical for financial data.

4. Select the cheapest available model.

Explanation

Incorrect: Low-cost models may compromise on accuracy for specialized tasks.

Overall explanation

Models optimized for numerical reasoning ensure accurate and meaningful summaries of data-rich financial reports.

Question 40

A Generative AI Engineer is tasked with building a guardrail for an LLM used in customer feedback analysis. The system must block responses containing offensive language. What implementation is most effective?

1. Train a classifier to flag offensive responses after generation.

Explanation

Incorrect: Post-generation flagging is less efficient than real-time filtering.

2. Integrate content filtering to detect and block offensive language in outputs.

Explanation

Correct: Content filtering ensures that offensive outputs are blocked, maintaining system reliability and user trust.

3. Fine-tune the LLM on a dataset with neutral language.

Explanation

Incorrect: Fine-tuning helps improve output tone but does not guarantee filtering offensive responses.

4. Reduce the model's temperature to minimize random outputs.

Explanation

Incorrect: Lowering temperature reduces randomness but does not directly block offensive language.

Overall explanation

Content filtering ensures offensive language is blocked in real-time, maintaining user trust and compliance with ethical standards.

Question 41

A Generative AI Engineer is designing a prompt for generating JSON-based responses from user queries about inventory details. Each response should include fields for item name, stock level, and location. Which prompt will achieve this?

1. Include details about the stock, but do not enforce a format.

Explanation

This choice suggests including details about the stock but does not enforce a specific format or mention the inclusion of item name, stock level, and location fields. It does not align with the prompt's requirement for generating JSON-based responses with specific fields.

2. Provide information about the inventory.

Explanation

This choice does not specify the inclusion of item name, stock level, and location fields in the response. It is too general and does not meet the specific requirements outlined in the prompt for generating JSON-based responses with specific fields.

3. Summarize the inventory details based on the query.

Explanation

This choice mentions summarizing inventory details based on the query, but it does not specify the inclusion of item name, stock level, and location fields in the response. It lacks the specificity required to meet the prompt's requirements for generating JSON-based responses with specific fields.

4. Generate a JSON response including item_name, stock_level, and location based on the query.

Explanation

This choice directly addresses the requirement of generating JSON-based responses with specific fields such as item name, stock level, and location. It ensures that the response format is consistent and includes all the necessary information as requested in the prompt.

Overall explanation

A well-structured prompt ensures responses are formatted consistently in JSON and include all required fields.

Question 42

A Generative AI Engineer must evaluate the safety of responses generated by a medical chatbot. What qualitative metric is most critical?

1. Stylistic tone for better engagement.

Explanation

Incorrect: While tone is helpful, accuracy is paramount in medical contexts.

2. Response speed to improve user experience.

Explanation

Incorrect: Speed is important but secondary to accuracy in medical applications.

3. Factual accuracy to ensure medically correct responses.

Explanation

Correct: Factual accuracy ensures the chatbot provides safe and reliable medical advice.

4. Length of the response for user readability.

Explanation

Incorrect: Length alone does not determine the quality or safety of a response.

Overall explanation

Ensuring factual accuracy is critical for generating safe and reliable responses in medical applications.

Question 43

A Generative AI Engineer must prevent hallucinations in an LLM's responses to medical queries. What is the most effective technique?

1. Train the model on diverse datasets.

Explanation

Incorrect: Training diversity does not guarantee factual accuracy in responses.

2. Ground the LLM's responses using a retrieval-augmented generation (RAG) pipeline connected to verified medical sources.

Explanation

Correct: RAG pipelines anchor LLM outputs in factual, retrieved data, reducing hallucinations.

3. Lower the temperature setting during inference.

Explanation

Incorrect: Lowering the temperature reduces randomness but does not eliminate hallucinations.

4. Use shorter prompts to limit response complexity.

Explanation

Incorrect: Shorter prompts do not directly impact the model's hallucination tendencies.

Overall explanation

A RAG pipeline grounds responses in factual data, ensuring outputs are accurate and reducing hallucinations in critical domains like medicine.

Question 44

A Generative AI Engineer must create a LangChain chain for generating personalized responses to user queries. The chain must include context from a vector store and generate coherent answers. What components should the chain include? **(Choose two)**

1. A rule-based system to hardcode responses.

Explanation

Incorrect: Rule-based systems lack flexibility for dynamic queries.

2. A classification model to label user intents.

Explanation

Incorrect: While useful for intent detection, classification does not directly contribute to context retrieval or response generation.

3. A summarization model for condensing user inputs.

Explanation

Incorrect: Summarization is not necessary for generating coherent answers.

4. A prompt template to structure user queries for the LLM.

Explanation

Correct: Prompt templates guide the LLM in generating coherent and relevant responses.

5. A retriever for fetching relevant context from the vector store.

Explanation

Correct: Retrievers enable dynamic access to context, improving response quality.

Overall explanation

Combining retrievers and prompt templates ensures dynamic and contextually relevant response generation using LangChain.

Question 45

A Generative AI Engineer must build a pipeline that converts unstructured legal documents into structured summaries. The summaries should highlight key clauses, risks, and obligations. Which tools should be included in the pipeline?

1. An LLM fine-tuned on legal text for generating structured summaries.

Explanation

Correct: Fine-tuning on legal data ensures the LLM produces domain-specific summaries with required elements.

2. A rule-based system for extracting clauses.

Explanation

Incorrect: Rule-based systems lack flexibility and adaptability for diverse legal texts.

3. A retrieval system for fetching similar legal cases.

Explanation

Incorrect: Retrieval alone does not structure or summarize content effectively.

4. A summarization model trained on general text.

Explanation

Incorrect: General models may not capture domain-specific nuances.

Overall explanation

Fine-tuned LLMs on legal data ensure the system generates structured summaries tailored to domain-specific requirements.